ELECTROIMPACT, INC.

Industry Sponsored Capstone Project Proposal

Aerospace Fastener Material Verification

Inline Aerospace Fastener Material Verification Using Eddy Current Sensing

Executive Summary

Develop a proof-of-concept inline sensing system capable of differentiating titanium and Inconel aerospace fasteners while they travel through a representative feed tube. The project builds on preliminary feasibility work indicating distinguishable eddy-current signatures between fastener materials.

Engineering Challenge

Design a sensor, electronics, and software platform that can reliably classify fastener material at production-relevant speeds while accounting for geometry changes, noise, and manufacturing variability.

Business Value

Automated material verification could reduce installation errors, improve quality assurance, reduce rework costs, and support future aerospace assembly automation initiatives.

Project Objectives

- Sensor and fixture design
- Signal acquisition and conditioning
- Material classification algorithms
- Performance characterization
- Industrial implementation recommendations

Student Skill Requirements

Electrical Engineering (electromagnetics, analog electronics), Computer Engineering (embedded systems, data acquisition, signal processing), Mechanical Engineering (fixture and automation integration).

Technology Readiness Roadmap

Current State: laboratory proof-of-concept. Capstone Deliverable: validated engineering prototype. Future Work: integration into automated aerospace feed systems.

Resources and Budget

Estimated prototype materials budget: $1,000–$3,000. Electroimpact to provide sample fasteners, representative tubing, engineering mentorship, and access to fabrication resources.

Mentor Expectations

Monthly technical reviews, requirements support, design feedback, hardware access, and participation in final evaluation.

Risks and Mitigation

Potential signal overlap, feed-speed limitations, and mechanical variability. Mitigation approaches include multi-frequency sensing, signal feature extraction, and repeatability testing.

Stretch Goals

Real-time embedded classification, multiple fastener family identification, machine-learning approaches, and representative automation integration.

Proposed Schedule

Phase	Semester	Deliverables
Requirements & Research	Fall	Requirements document, test plan
Sensor Design	Fall	Prototype design package
Prototype Build	Fall	Functional sensing fixture
Data Collection	Spring	Experimental dataset
Optimization & Validation	Spring	Classification results
Final Delivery	Spring	Report, presentation, demonstration